

PATRICK SEAMUS ROONEY

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EDUCATION

Virginia Tech– Blacksburg, VA

Bachelor of Science– Computer Engineering

Expected May 2027

GPA 3.44

Admitted to Tech Talent Accelerated Master's Program in Computer Engineering

Minor in Mathematics and Cybersecurity

Relevant Courses: Embedded Systems, Computer Networking, Cryptography, Data Structures and Algorithms, Signals and Systems, Physical Electronics, Digital systems, Computational Engineering, Circuits and Devices

PROJECTS EXPERIENCE

Personal Smart Locker, Integrated Design Project

August 2025 - December 2025

- Built a standalone smart-locker system using an ATmega328p, servo motor, keypad, and voltage-regulation circuitry.
- Features include: encrypted password storage, password change logic, low-power modes, motor-blockage detection, low-battery monitoring/alerts, a buzzer UI, and MOSFET based switching for battery efficiency.

16-bit CPU, Digital Systems

August 2024 - December 2024

- Designed a 16-bit CPU in SystemVerilog featuring a custom bus system, registers, ALU, and 19 opcode instructions.
- Optimized transistor count and minimized propagation delay to improve overall performance by using Karnaugh Maps.

WORK/LEADERSHIP EXPERIENCE

Full Stack Software Development Intern

August 2025 - Present

Counter– Hybrid / Newark, NJ

- Developing a machine learning model to predict SSDI claim outcomes using real-world client data. Built scoring criteria and mapped the criteria to Supabase data. Currently researching embedding and tabular models for implementation.
- Test and update existing application to ensure reliability and performance.
- Integrating Supabase datasets into existing application to enable interactive analytics and enhance user insights.
- Implemented and tested APIs to streamline Counter's SSDI claims workflow, enhancing reliability and reducing manual processing time.

Vice President

December 2023 - Present

Virginia Tech Club Wrestling– Blacksburg, VA

- Entrusted in leading and managing a competitive collegiate wrestling club, overseeing team operations, member development of 50+ student-athletes, organizing practices, working with Division 1 wrestling program, leading community service events, and all other organizational responsibilities with the President and Coach.

Head Of Leadership Team

August 2020 - April 2023

West Morris Soccer Club TOPSoccer– Mendham, NJ

- Organized weekly soccer/basketball events for children with special needs and lead a team of 50+ volunteers and 70+ special needs players.
- Co-founder and leader of TOPZoom, a virtual alternative to TOPSoccer. Created curriculum, protocol, training, and elicited parent feedback. Created a 50-page TOPZoom policy manual. Presented at TOPSoccer National Symposium about TOPZoom.

Co-Founder/Camper Relations Team Lead

April 2020 - August 2021

BuddEConnect– Chester, NJ

- Co-founded BuddEConnect, a virtual, nonprofit camp for children with special needs.
- Responsible for recruitment and management of relations of 50+ campers and families in 2020 and 60+ in 2021 from 19 different states. Facilitated a \$15,000 single donor challenger grant.

SKILLS

Computer: Machine Learning, Application Development, API Development, Computer Networks, Embedded Systems, Microcontrollers, C++, C, Python, Assembly, JavaScript, Microsoft Office Suite

Technical: Circuit Design, Filter Design, Cryptography, Digital Design, Troubleshooting, Debugging